Lifting minimal covers of an interval

Private continuation, phase 277, 11 October 2026. The rank-four theorem used below is the accepted theorem of phase 256, independently reviewed in phase 273 and published in phase 274. Nothing here proves finite-cover multiplicity when the only intersection defect has rank at least five.

1. The extension supplied by the finite-cover construction

Let $G = \langle g_0, \dots, g_{n-1} \rangle$ be a finite string group. All maps in this note preserve the ordered marks. For a contiguous interval $I = [i, j]$, write $G_I = \langle g_i, \dots, g_j \rangle$.

Lemma 1. Let $Q \to G_{[i,j]}$ be a finite string C-group cover. There is a finite string C-group cover $R \to G$ whose marked subgroup $R_{[i,j]} \to G_{[i,j]}$ is isomorphic to the given cover.

Proof. We spell out the preservation assertion in the construction of Monson–Schulte, Lemma 3.3, rather than use only its finite-existence conclusion. Suppose first that $i = 0$ and $j < n - 1$. A finite regular extension E of Q, supplied by their Lemma 3.1, has an initial maximal parabolic identified with Q. Form the subgroup

$$R' = \langle (g_t, e_t) : 0 \le t \le j + 1 \rangle \le G_{[0,j+1]} \times E.$$

The initial subgroup generated by the first $j + 1$ marks is the graph of the epimorphism $Q \to G_{[0,j]}$, and is therefore exactly a copy of Q. Projection to E is an epimorphism and is injective on that maximal parabolic. The standard quotient criterion proves that R' is a string C-group. Projection to the first factor covers $G_{[0,j+1]}$, and both factors are finite. This is the actual proof of Monson–Schulte's lemma. In particular it preserves the given cover Q, not merely those smaller parabolics on which its map happens to be injective.

Starting from an arbitrary interval $[i, j]$, apply the same construction to the consecutively relabelled string $g_i, \dots, g_{j+1}$. Repeat to extend the right endpoint up to $n - 1$. Then reverse the order of the marks and apply the construction repeatedly to extend the left endpoint down to zero. Each extension preserves the previously constructed maximal parabolic exactly, and therefore preserves its subgroup on the original interval. The composite epimorphisms preserve all marks. ■

The classical input is the seven-page author version of Monson–Schulte: Lemma 3.1 on PDF pages 4–5, and the direct-product construction in the proof of Lemma 3.3 on PDF page 5. Its Theorem 3.4 gives existence but, by itself, would conceal the preservation assertion needed here. The source bytes are bound in phase277_sources.json. This is a use of prior mathematics, not a new finite-cover existence claim.

2. Minimal descendants retain a minimal interval

A finite string C-group cover $Q \to L$ is minimal if every string C-group quotient still covering L is the same marked cover.

Theorem 2. Every finite minimal string C-group cover of a contiguous parabolic G_I occurs as the I-parabolic of some finite minimal string C-group cover of G.

Proof. Begin with a minimal cover $Q \to G_I$ and the extension $R \to G$ from Lemma 1. Among the string C-group quotients of R still mapping onto G, choose a minimal one, denoted M. This is possible because R is finite. Every further C-group quotient of M over G would be a quotient of R, so M is minimal among all C-covers of G, not just within a smaller list.

The quotient map restricts to a surjective marked homomorphism

$$Q \cong R_I \longrightarrow M_I \longrightarrow G_I.$$

A standard parabolic of a string C-group is a string C-group. Thus M_I is a C-group quotient of Q still covering G_I. Minimality of Q forces the first map to be an isomorphism. ■

Corollary 3. If G_I has at least m pairwise nonisomorphic finite minimal C-covers, then G has at least m, for every positive integer m. If G_I has infinitely many, so does G.

Proof. Apply Theorem 2 to each local cover. A marked isomorphism of two resulting global groups restricts to a marked isomorphism of their I-parabolics. Distinct local covers therefore produce distinct global ones. The same argument works for every finite subfamily of an infinite family. ■

This does not assert that the parabolic of an arbitrary global minimal cover must be minimal. The theorem constructs global minimal covers whose specified parabolic is the prescribed local minimal one.

3. A consequence of the rank-four theorem in every rank

Corollary 4. Let P be any finite polytope, of arbitrary rank, and let $G = \mathrm{Mon}(P)$. If some subgroup generated by four consecutive marks is not a string C-group, then P has infinitely many finite minimal regular polytopal covers.

Proof. Let $I = [i, i+3]$ be such an interval. Each endpoint mark lies outside the group generated by the other three: it lies outside the group generated by all other marks of G, by the flag-face argument. The four marks are distinct nonidentity involutions and satisfy the required string relations. Hence Theorem 1 of phase 256 applies directly to the finite marked group G_I; no separate assertion that G_I is itself a polytope's monodromy group is needed.

For every prime $p > 4|G_I|$, that theorem gives a finite C-cover with p-group kernel. A minimal C-group quotient still covering G_I has nontrivial p-power kernel, since G_I is not C. These local minimal covers have distinct orders for distinct primes. Corollary 3 now gives infinitely many global finite minimal C-covers of G. The usual normal flag-kernel correspondence turns them into finite minimal regular polytopal covers of P. ■

No bound or prime-power assertion about the *global* cover degree is made. The extension procedure can introduce other primes. Distinctness is read from the indicated interval parabolic, not inferred from the total orders.

More generally, any later finite-multiplicity theorem for a marked interval can be transported by Theorem 2 without repeating this argument.

4. The genuinely unresolved case

Independent rank-three string groups need not be C-groups. The familiar rank-three monodromy theorem is a statement about actual polytopes and cannot be applied to this broader class. There is a direct reduction to phase 256 which supplies the needed rank-three result instead. For a finite independent rank-three group $L = \langle a, b, c\rangle$, append an independent central involution t, obtaining $L \times C_2$ with marks $(a,1), (b,1), (c,1), (1,t)$. Both extreme independence conditions of phase 256 hold. Restrict each large-prime p-kernel C-cover supplied by phase 256 to its first three marks. This parabolic is a C-group mapping onto L, with a p-group kernel. If L is non-C, its finite minimal descendants have nontrivial p-power degrees, and distinct primes give infinitely many. Ranks one and two already satisfy the intersection property.

Now suppose a finite non-C string group with independent marks has only finitely many finite minimal C-covers, and choose such an example of least rank. All its proper contiguous parabolics must be C-groups. Otherwise a proper non-C interval has infinitely many finite minimal covers by minimality of the rank, and Corollary 3 contradicts finiteness for the whole group. Thus the smallest unresolved rank for this stronger infinitude statement is at least five, and the obstruction may be assumed to be the top intersection

$$G_{[0,n-2]} \cap G_{[1,n-1]} \neq G_{[1,n-2]},$$

with both end groups already C. This is a reduction in the broader class of independent marked groups; it should not silently identify every such group with an actual polytope monodromy group.

The analogous conclusion from a least-rank counterexample restricted only to actual polytopes needs care: a parabolic of their monodromy group acts on the disjoint union of many sections and need not have been separately identified as the monodromy group of one polytope. Corollary 4 avoids that issue because phase 256 is a theorem for the broader marked-group class.

The locally faithful amalgam criterion in the previously reviewed phase 246 manuscript remains relevant to this remaining case. It does not by itself suffice: the Tomotope already shows that locally faithful odd-primary covers may not exist. Rank four was resolved by enlarging the end groups before the second homology cover. A higher-dimensional substitute for that end enlargement, with simultaneous control of all required intersections, is the outstanding mathematical step.